Some remarks about machine learning and (un)natural proving

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Malinca workshop
October 3, 2025, IHP, Paris

European Research Council
Established by the European Commission

What is this talk about?

- I have given many propaganda talks about ML/AI/TP/etc recently
- one of them last year in Paris
  `https://josefurban.eu/inria24.pdf`
- and Adrian has already talked about our current autoformalization project with Atle
- I have become also a bit of a contrarian re (naive) claims about superiority of DL/LM in TP
- So these are several (random?) remarks that may be relevant to the users of ML and human-level proving

Some more pointers

- My 2014 talk from IHP - `https://www.ciirc.cvut.cz/~urbanjo3/ihp14/ihp14.html`
- declared three AITP challenges, one about autoformalization
- the first decade of autoformalization of talk in Bonn - `https://www.youtube.com/watch?v=4JeezEGc_gQ`
- a larger overview of learning-guided ATP in 2024 - `https://arxiv.org/abs/2403.04017`

AITP Challenges/Bets from 2014

- In 20 years, 80% of Mizar and Flyspeck toplevel theorems will be provable automatically (same hardware, same libraries as in 2014 - about 40% then)
- In 10 years: 60% (DONE already in 2021 - 3 years ahead of schedule)
- In 25 years, 50% of the toplevel statements in LaTeX-written Msc-level math curriculum textbooks will be parsed automatically and with correct formal semantics (this may be faster than I expected)

My original plan (1996): learn doing math and theorem proving like humans do - from many textbooks

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1 of 23 /home/urban/ANGL.DVI MScThesisPaper.pdf 191.8%

First the notion of the heuristic and its importance for automatic theorem proving is explained. The question of how to construct a useful system of heuristics and notions is posed, and it is argued for the importance of learning. A simple example of learning a lemma is shown using the Inductive Logic Programming system Progol.

Second a general framework for learning heuristics is discussed and the proof checking system Mizar providing a large body of mathematics is found suitable to learn upon. The process of modifying Mizar for use with Progol is shortly described, e.g. turning the mathematical texts to Prolog structures, creating initial notions, creating the initial base of rules from theorems and definitions, preparing proofs for learning, and problems with polymorphism.

Finally first experiments in this environment are described, which include learning the notion of atomic formula and learning how to use initial rules in the proof of Rolle's theorem.

My original plan

- related to Malinca?
- I wanted more human theorem proving and math development methods than what the ATPs were offering
- they were encountering combinatorial explosion
- weren't reasoning with a lot of heuristic/crisp (problem-solving) knowledge like we do
- the Cyc project by Doug Lenat tried to invent such heuristics manually
- but learning looked like a useful meta heuristic

First Remark - ATP and its History

- In Adrian's talk we heard that ATPs are good in Tarski's geometry
- This is not totally true
- Beeson and Wos had to cheat to get some of the harder proofs with Otter
- However this relates to some core ideas present from the beginning of ATP

Martin Davis: Early History of Automated Deduction

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5 (2 of 12) The Early History of Automated Deduction: Dedicated to the memory of Hao Wang
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With the ready availability of serious computer power, deductive reasoning, especially as embodied in mathematics, presented an ideal target for those interested in experimenting with computer programs that purported to implement the "higher" human faculties. This was because mathematical reasoning combines objectivity with creativity in a way difficult to find in other domains. For this endeavor, two paths presented themselves. One was to try to understand what people do when they create proofs and to write programs emulating that process. The other was to make use of the systematic work of the logicians in reducing logical reasoning to standard canonical forms on which algorithms could be based. Each path confronted daunting obstacles. The difficulty with the first approach was that available information about how creative mathematicians go about their business was and remains vague and anecdotal. On the other hand, the well-known unsolvability results of Church and Turing showed that the kind of algorithm on which a programmer might want to base a theorem-proving program simply did not exist. Moreover, it was all too obvious that an attempt to generate a proof of something non-trivial by beginning with the axioms of some logical system and systematically applying the rules of inference in all possible directions was sure to lead to a gigantic combinatorial explosion.

Martin Davis: Early History of Automated Deduction

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With the ready availability of serious computer power, deductive reasoning, especially as embodied in mathematics, presented an ideal target for those interested in experimenting with computer programs that purported to implement the "higher" human faculties. This was because mathematical reasoning combines objectivity with creativity in a way difficult to find in other domains. For this endeavor, two paths presented themselves. One was to try to understand what people do when they create proofs and to write programs emulating that process. The other was to make use of the systematic work of the logicians in reducing logical reasoning to standard canonical forms on which algorithms could be based. Each path confronted daunting obstacles. The difficulty with the first approach was that available information about how creative mathematicians go about their business was and remains vague and anecdotal. On the other hand, the well-known unsolvability results of Church and Turing showed that the kind of algorithm on which a programmer might want to base a theorem-proving program simply did not exist. Moreover, it was all too obvious that an attempt to generate a proof of something non-trivial by beginning with the axioms of some logical system and systematically applying the rules of inference in all possible directions was sure to lead to a gigantic combinatorial explosion.

Martin Davis: Early History of Automated Deduction

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versity was attended by almost every logician working in the United States. Many of the more theoretically inclined researchers from the nearby IBM facilities were also present; FORTRAN, a brand-new innovation in programming practice was unveiled. After discussions with Gelernter, the logician Abraham Robinson was led to give a short talk [Robinson 1957] in which he pointed to Skolem functions and "Herbrand's theorem" as useful tools for general purpose theorem-provers. He also made the provocative remark that the auxilliary points, lines, or circles "constructed" as

part of the solution to a geometry problem can be thought of as being elements of what is now called the Herbrand universe for the problem.

The first theorem-provers for first-order logic to be implemented based on Herbrand's theorem employed a completely unguided search of the Herbrand universe. Instead of using Skolem functions to deal with instantiations, variables were re-

Martin Davis (Logicists vs AI) Today

- in general a lot of symmetry breaking in ATPs (e.g. orderings in Assia's talk)
- and even learning in SAT solvers (the CDCL: revolution!)
- it's unclear how these things can/should/will interact as we get more ML
- a lot of ideas and experiments starting with the Munich group (Schulz, Denzinger, etc) in the 1990s
- many ways how to do ML and guide the TPs, some more and some less human

Martin Davis Today? - Learning instantiations

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Invariant neural architecture for learning term synthesis in instantiation proving
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Invariant neural architecture for learning term synthesis in instantiation proving

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ARTICLE INFO

Article history:
Received 18 September 2023
Received in revised form 14 August 2024
Accepted 23 August 2024
Available online 28 August 2024

Keywords:
Automated theorem proving
Graph neural networks
Instantiation

ABSTRACT

The development of strong CDCL-based propositional (SAT) solvers has greatly advanced several areas of automated reasoning (AR). One of the directions in AR is therefore to make use of SAT solvers in expressive formalisms such as first-order logic, for which large corpora of general mathematical problems exist today. This is possible due to Herbrand's theorem, which allows reduction of first-order problems to propositional problems by instantiation. The core challenge is synthesizing the appropriate instances from the typically infinite Herbrand universe.

In this work, we develop a machine learning system targeting this task, addressing its combinatorial and invariance properties. In particular, we develop a GNN2RNN architecture based on a graph neural network (GNN) that learns from problems and their solutions independently of many symmetries and symbol names (addressing the abundance of Skolems), combined with a recurrent neural network (RNN) that proposes for each clause its instantiations. The architecture is then combined with an efficient ground solver and, starting with zero knowledge, iteratively trained on a large corpus of mathematical problems. We show that the system is capable of solving many problems by such educated guessing, finding proofs for 32.12% of the training set. The final trained system solves 19.74% of the unseen test data on its own.

Martin Davis Today? - Learning instantiations

- SAT solvers are today superhuman
- istantiation-based ATP has become very strong recently (iprover, SMTs)
- can we combine guessing of instances with SAT solving?
- Jelle Piepenbrock's GNN2RNN architecture
- the invariant/symbol-independent architecture generalizes here (much) better here than the transformer
- might be also doing more efficient "reasoning" than a transformer model of the same size
- we have not tried heavy chain of thought reasoning models here, just our standard learning-proving feedback loop
- intelligent (symmetry breaking) decision procedures like ground congruence closure help a lot - much less guessing needed (we don't need the equality axioms)
- will this ever work with pure LLMs or do we need "reasoning" models?

Feedback loops between thinking and learning

- Science and Method: Ideas about the interplay between correct deduction and induction/intuition
- *"And in demonstration itself logic is not all. The true mathematical reasoning is a real induction [...]"*
- I believe he was right: strong general reasoning engines have to combine deduction and induction (learning patterns from data, making conjectures, etc.)

Feedback loops between thinking and learning

- I guess I always planned to build bigger proving-learning metasystems
- But it was a bit of a surprise and a very interesting object when I did the first one
- first in Malarea (2006) - I had to debug my MPTP translation because I didn't want to learn from buggy proofs
- btw this got exposed on Isabelle problems in 2007:
- Malarea solved all of them very quickly, exploiting the (intentional) unsoundness of the translation
- does this have any implications today for LLM-only (no formal proof) RL (self-learning) over math?

Feedback loops between thinking and learning

- the Malarea feedback loop has been reincarnated in various settings:
- strategy evolution (Blind Strategymaker and its descendents - 2012)
- internal guidance of simple connection tableau provers (leanCoP ==> rlCoP - 2018)
- internal guidance of state-of-the art ATPs (ENIGMA - 2019, Deepire, etc)
- learning guided tactical provers (Tactictoe - 2017 and its followers)
- instantiation provers (iprover, CVC5 - 2023)
- etc

ENIGMA Prove/Learn feedback loop on MML

- Done on 57880 Mizar Mathematical Library formal math problems in 2019
- Efficient ML-guidance inside the best ATPs like E prover (ENIGMA)
- *Training* of the ML-guidance is *interleaved* with *proving* harder problems
- Ultimately a 70% improvement over the original E strategy:
- ... from 14933 proofs to 25397 proofs (all in 10s CPU - no cheating)

	$\mathcal{S}$	$\mathcal{S} \odot \mathcal{M}_9^0$	$\mathcal{S} \oplus \mathcal{M}_9^0$	$\mathcal{S} \odot \mathcal{M}_9^1$	$\mathcal{S} \oplus \mathcal{M}_9^1$	$\mathcal{S} \odot \mathcal{M}_9^2$	$\mathcal{S} \oplus \mathcal{M}_9^2$	$\mathcal{S} \odot \mathcal{M}_9^3$	$\mathcal{S} \oplus \mathcal{M}_9^3$
solved	**14933**	16574	20366	21564	22839	22413	23467	22910	23753
$\mathcal{S}\%$	+0%	+10.5%	+35.8%	+43.8%	+52.3%	+49.4%	+56.5%	+52.8%	+58.4
$\mathcal{S}+$	+0	+4364	+6215	+7774	+8414	+8407	+8964	+8822	+9274
$\mathcal{S}-$	-0	-2723	-782	-1143	-508	-927	-430	-845	-454

	$\mathcal{S} \odot \mathcal{M}_{12}^3$	$\mathcal{S} \oplus \mathcal{M}_{12}^3$	$\mathcal{S} \odot \mathcal{M}_{16}^3$	$\mathcal{S} \oplus \mathcal{M}_{16}^3$
solved	24159	24701	25100	**25397**
$\mathcal{S}\%$	+61.1%	+64.8%	+68.0%	**+70.0%**
$\mathcal{S}+$	+9761	+10063	+10476	+10647
$\mathcal{S}-$	-535	-295	-309	-183

- 75% of the Mizar corpus (43414) reached in July 2021 - higher times and many prove/learn cycles: `https://github.com/ai4reason/ATP_Proofs`
- Details in our Mizar60 paper: `https://arxiv.org/abs/2303.06686`

ENIGMA - The Rise of Computronium (2021)

From: Josef Urban <josef.urban@gmail.com>, Date: Jul 26, 2021 at 9:47
Subject: ENIGMA - The Rise of Computronium

I am happy to announce that the ENIGMA system of the E lineage, helped by its Deepire Vampiric cousin, has reached today (July 26th, 2021) the landmark of 75% automatically proved Mizar top-level problems.
[..]
Many of the proofs show that ENIGMA has autonomously (without any human-programmed decision procedures, tactics, and/or dataset preparation/tuning) learned how to routinely perform common mathematical algorithmic tasks such as numeric calculation, matrix manipulation, boolean algebra, integration and differentiation, sequences of standard rewriting and normalization operations in various algebraic theories, etc., and combine them with other reasoning tasks needed for completing the fully formal proofs.
[..]
This suggests that learned guidance combined with efficient search may in near future lead to a new fully declarative problem-solving computing/reasoning architectures applicable to arbitrary computing/reasoning problems without any human engineering.
[..]
Combined with autoformalization this could lead us to large deployment of reasoning machines in science, following Leibniz's Calculemus dream.

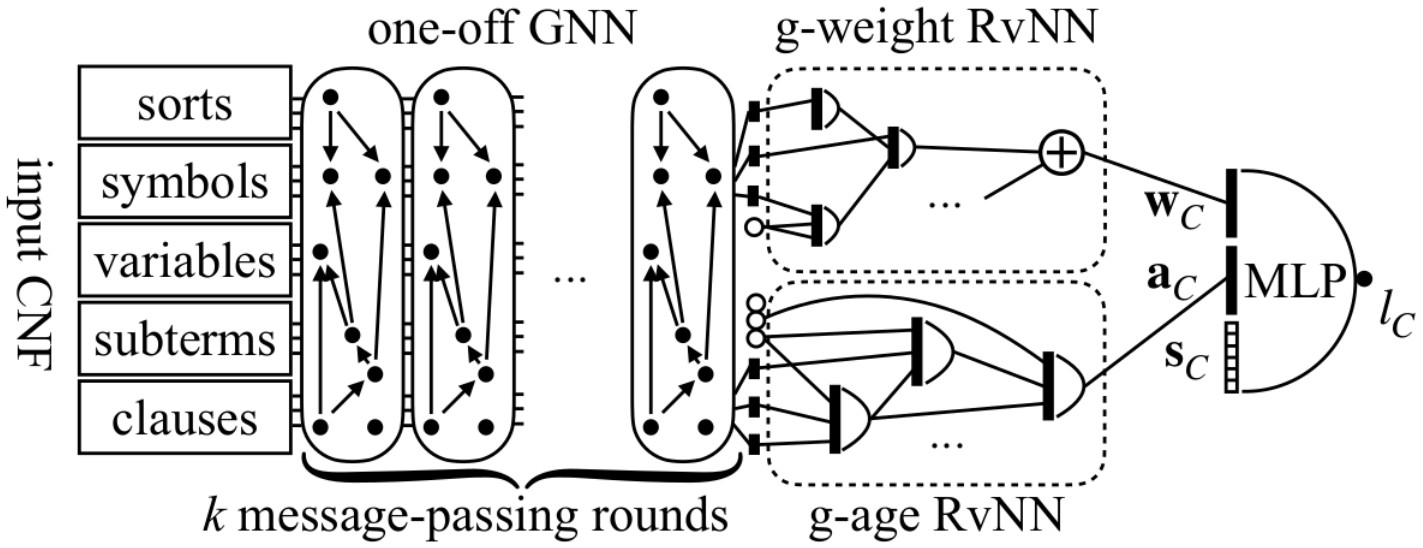
More Low-Level Guidance of Various Creatures

- Neural (TNN) clause selection in Vampire (Deepire - M. Suda):
  Learn from clause *derivation trees only*
  Not looking at what it says, just who its ancestors were.
- Fast and surprisingly good: Extreme Deepire/AVATAR proof of
  $\epsilon_0 = \omega^{\omega^{\omega^{\cdot^{\cdot^{\cdot}}}}}$ `https://bit.ly/3Ne4WNX`
- 1193-long proof takes *about the same resources as one GPT-3/4 reply*
- GNN-based guidance in iProver (Chvalovsky, Korovin, Piepenbrock)
- New (*dynamic data*) way of training
- Led to doubled real-time performance of iProver's instantiation mode
- CVC5: neural & GBDT instantiation guidance (Piepenbrock, Jakubuv)
- very recently 20% improvement on Mizar
- Hints method for Otter/Prover9 (Veroff):
- boost inferences on clauses that match a lemma used in a related proof
- symbolic ML - can be combined with statistical - proof completion vectors

Latest Deepire - Martin Suda

- 20% improvement over the best ATP!
- After improving it by 27% in CASC last year!
- Everything is done in real time on a single CPU
- only 30% slowdown by the NN guidance compared to unguided Vampire
- very good engineering and tight integration of the NNs
- no cheating by hiding the remote GPU cloud
- solved several unknown problems in TPTP and proved many previously unproved
- Could LLMs do it? If yes, why have they not done it yet? If yes, with what resources?

Latest Deepire - Martin Suda



. Clause evaluation neural architecture. Information flows from left to right.

The QED dream vs the LLM dream

The QED Manifesto
May 15, 1994
QED is the very tentative title of a project to build a computer system that effectively represents all important mathematical knowledge and techniques.

The QED Manifesto

The development of mathematics toward greater precision has led, as is well known, to the formalization of large tracts of it, so that one can prove any theorem using nothing but a few mechanical rules.
– K. Godel
If civilization continues to advance, in the next two thousand years the overwhelming novelty in human thought will be the dominance of mathematical understanding.
– A. N. Whitehead

The QED Manifesto

The idea is to make a language such that everything we write in it is interpretable as correct mathematics ... This may include the writing of a vast mathematical encyclopedia, to which everybody (either a human or a machine) may contribute what he likes. The idea of a kind of formalized encyclopedia was already conceived and partly carried out by Peano around 1900, but that was still far from what we might call automatically readable.
– N. G. de Bruijn

The QED dream vs the AI/LLM dream

Project QED is largely independent of the question of the possibility or utility of artificial intelligence or automated reasoning. To the extent that mechanical aids of any kind can be used to help construct (or shorten) entries in the QED system, we can be appreciative of such aids, even if the aids use techniques that are from the realms of artificial intelligence, assuming of course that what the aids suggest doing is verifiably correct. A key fact is that it will not matter, from the viewpoint of soundness, whether proofs were added to the QED system by humans, dumb programs, smart programs or some combination thereof. All of the QED system will be checkable by a simple program, from first principles.

Some analogies between the LLMs and ATPs

Learning-guided ATPs as a more logical version of LLMs?

`https://github.com/JUrban/LMvsATP`

Declarative proofs

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Stanisław Jaśkowski

9 languages

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From Wikipedia, the free encyclopedia

Stanisław Jaśkowski (Polish pronunciation: [staɲsvaf jaɕkɔfskʲi]; 22 April 1906, in Warsaw – 16 November 1965, in Warsaw) was a Polish logician who made important contributions to proof theory and formal semantics. He was a student of Jan Łukasiewicz and a member of the Lwów–Warsaw School of Logic. He is regarded as one of the founders of natural deduction, which he discovered independently of Gerhard Gentzen in the 1930s.[1] He is also known for his research into paraconsistent logic.[2] Upon his death, his name was added to the Genius Wall of Fame. He was the President (rector) of the Nicolaus Copernicus University in Toruń.

Stanisław Jaśkowski

Born	22 April 1906 Warsaw, Congress Poland
Died	16 November 1965 (aged 59) Warsaw, Poland
Nationality	Polish

Life and career [edit]

He was born in 1906 in Warsaw to father Feliks Jaśkowski and mother Kazimiera (nee Dzierzbicka). In 1924, he graduated from high school in Zakopane and enrolled at the University of Warsaw to study mathematics. He was taught mathematical logic under Jan Łukasiewicz and participated in the Polish Mathematicians' Congresses in Lviv (1927) and Vilnius (1931).

Declarative proofs

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jaskowski.pdf 4 of 31

§ 1. The theory of deduction based on the method of suppositions.

In 1926 Prof. J. Ł u k a s i e w i c z called attention to the fact that mathematicians in their proofs do not appeal to the theses of the theory of deduction, but make use of other methods of reasoning. The chief means employed in their method ist that of an arbitrary supposition. The problem raised by Mr. Ł u-k a s i e w i c z was to put these methods unter the form of structural rules and to analyze their relation to the theory of deduction. The present paper contains the solution of that problem.[1])

Here we consider as structural rules only those which re-fer to the external appearance of expressions. It is possible to formulate such rules only for a formal system in which all pro-positions are written in symbols. In the present paper, we shall use Mr. Ł u k a s i e w i c z's [2]) bracket - free symbolism. The implication „if α, then β" will be symbolized by „C$\alpha\beta$" and the negation „not α" by „Nα". In the above, α und β stand for significant expressions of the system. The significant

Declarative proofs

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[1]) The first results on that subject obtained by the author in 1926 at Prof. Ł u k a s i e w i c z's seminar were presented at the First Polish Mathematical Congres in Lwów in 1927 and were mentioned in the proceedings of the Congress: *Kalçga pamiqtkowa pierwszego polskiego zjazdu matematycznego*, Kraków 1929.

J. Ł u k a s i e w i c z, *Elementy logiki matematycznej*. Opracował M. Presburger, Warszawa 1929 (litogr.) p. 38 *ss*. - J. Ł u k a s i e w i c z und A. Tarksi. *Untersuchungen über den Aussagenkalkül*. Comptes Rendus des séances de la Soc. des Sciences et des Lettres de Varsovie, XXIII. 1930. Classe III.

Declarative proofs

- Jaskowski - on the rules of supposition (1926?)
- - picked up by the Mizar people in 1970s
- brought to the "West" in 1996 by John Harrison (Mizar mode for HOL Light) following the QED workshop in warsaw
- Isar in early 2000, Naproche, SAD, etc

Side note: Declarative proofs vs Discourse Representation Theory

- There may be a path from Jaskowski to DRT - maybe via Tarski and Montague
- The Mizar data structures (that hold local constants and local assumptions) vs
- the data structures in Naproches as presented by Daniel kuhlwein in 2011 (called DRT)

Declarative proofs vs ML

- are declarative proofs what we really want in ATP/AITP?
- people sometimes say that it may be good for (human?) math (all of it?)
- ... but perhaps not so much for SW/HW verification
- the Kimina prover talk at AITP:
- declarative Lean proofs easier for humans to debug rather than nondeclarative

Declarative proofs vs ML

- Can we do ML for declarative proofs? Is it useful?
- the Karpathy RNN experiment on Stacks from 2015 - impressive but gibberish
- my GPT2-over-Mizar experiment from 2020 - interesting but very non-semantic
- (after that all help broke lose with “pretrained” LLMs I excused myself)
- IMHO, having more generalizing architectures may improve on transformers
- some classical ATP people are quite skeptical:
- no symmetry breaking methods, etc, very unconstrained etc

Another way: Convert ATP proofs to ITP proofs?

- very recent experiments with Theo Stoskopf and Jonas Bayer (motivated by Cyril's talk in Goteborg)
- reconstruction of the Sledgehammer proofs mentioned by Makarius?
- Qwen 32B finetuned on 10k TPTP and Mizar proofs
- Theo makes them do long "reasoning" to get from the TPTP to the Mizar proof
- the model reasons quite a lot to get to the final Mizar proof

Another way: Convert ATP proofs to ITP proofs?

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```
    "source": "classes5",
    "name": "classes5__t8_classes5",
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 ![X1]:(~(v1_xboole_0(X1))=>r2_wellord2(X1,k11_eqrel_1(X1))), file('classes5/classes5__t8_classes5', t8_classes5)).\nfo
f(t5_card_1, axiom, ![X1, X2]:(r2_wellord2(X1,X2)<=>k1_card_1(X1)=k1_card_1(X2)), file('classes5/classes5__t8_classes5'
, t5_card_1)).\nfof(t12_topgen_2, axiom, ![X1]:k1_card_1(k11_eqrel_1(X1))=k1_card_1(X1), file('classes5/classes5__t8_cl
asses5', t12_topgen_2)).\nfof(c_0_3, negated_conjecture, ~(![X1]:(~v1_xboole_0(X1)=>r2_wellord2(X1,k11_eqrel_1(X1)))),
inference(fof_simplification,[status(thm)],[inference(assume_negation,[status(cth)],[t8_classes5])])).\nfof(c_0_4, nega
ted_conjecture, (~v1_xboole_0(esk1_0)&~r2_wellord2(esk1_0,k11_eqrel_1(esk1_0))), inference(skolemize,[status(esa)],[inf
erence(variable_rename,[status(thm)],[inference(fof_nnf,[status(thm)],[c_0_3])])])).\nfof(c_0_5, plain, ![X11, X12]:((~
r2_wellord2(X11,X12)|k1_card_1(X11)=k1_card_1(X12))&(k1_card_1(X11)!=k1_card_1(X12)|r2_wellord2(X11,X12))), inference(v
ariable_rename,[status(thm)],[inference(fof_nnf,[status(thm)],[t5_card_1])])).\nfof(c_0_6, plain, ![X10]:k1_card_1(k11_
eqrel_1(X10))=k1_card_1(X10), inference(variable_rename,[status(thm)],[t12_topgen_2])).\ncnf(c_0_7, negated_conjecture,
 (~r2_wellord2(esk1_0,k11_eqrel_1(esk1_0))), inference(split_conjunct,[status(thm)],[c_0_4])).\ncnf(c_0_8, plain, (r2_w
ellord2(X1,X2)|k1_card_1(X1)!=k1_card_1(X2)), inference(split_conjunct,[status(thm)],[c_0_5])).\ncnf(c_0_9, plain, (k1_
card_1(k11_eqrel_1(X1))=k1_card_1(X1)), inference(split_conjunct,[status(thm)],[c_0_6])).\ncnf(c_0_10, negated_conjectu
re, ($false), inference(cn,[status(thm)],[inference(rw,[status(thm)],[inference(spm,[status(thm)],[c_0_7, c_0_8]), c_0_
9])]), ['proof']).\n# SZS output end CNFRefutation",
    "formal_statement": "theorem\n  for X being non empty set holds X,SmallestPartition X are_equipotent",
    "statement": "theorem\n  for X being non empty set holds X,SmallestPartition X are_equipotent",
    "human_proof": "let X be non empty set;\n    card X = card SmallestPartition X by TOPGEN_2:12; \n    hence thesis b
y CARD_1:5;\n  end;",
    "content": "<think>\nStep 1. Reading the conjecture\nMy journey begins with the main theorem to be proven: `t8_clas
ses5`. It states `![X1]:(~(v1_xboole_0(X1))=>r2_wellord2(X1,k11_eqrel_1(X1)))`.\n\nLet's try to decipher this. `v1_xboo
le_0(X1)` is a property of a set `X1`. Given the context of \"non empty set\" in the human-readable statement, `~v1_xbo
ole_0(X1)` likely means \"X1 is not empty\".\nThe predicate `r2_wellord2(A, B)` seems to represent a relation between t
wo sets. The human statement \"are_equipotent\" is a strong hint. So, I'll hypothesize `r2_wellord2` means \"are equipo
tent\".\nThe function `k11_eqrel_1(X1)` takes a set `X1` and returns another object, which based on context, appears to
 be the \"Smallest Partition of X1\".\n\nSo, in plain English, the conjecture is: **For any non-empty set X1, X1 is equ
ipotent to its Smallest Partition.**\n\nStep 2. Examining the axioms for clues\nTo prove this, the machine used two axi
oms. Let's see what they tell us.\n\n*   **Axiom t5_card_1:** `![X1, X2]:(r2_wellord2(X1,X2)<=>k1_card_1(X1)=k1_card_1(
X2))`\n    This is a major breakthrough! It looks like `k1_card_1` is the cardinality of a set. The axiom states that t
:
```

Another way: Convert ATP proofs to ITP proofs?

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urban@dell: ~/Downloads/evalres1

junct,[status(thm)],[c_0_6])).\ncnf(c_0_10, negated_conjecture, ($false), inference(cn,[status(thm)],[inference(rw,[status(thm)],[inference(spm,[status(thm)],[c_0_7, c_0_8]), c_0_9])]), ['proof']).\n# SZS output end CNFRefutation",
    "formal_statement": "theorem\n for X being non empty set holds X,SmallestPartition X are_equipotent",
    "statement": "theorem\n for X being non empty set holds X,SmallestPartition X are_equipotent",
    "human_proof": "let X be non empty set;\n   card X = card SmallestPartition X by TOPGEN_2:12; \n   hence thesis by CARD_1:5;\n end;",
    "content": "<think>\nStep 1. Reading the conjecture\nMy journey begins with the main theorem to be proven: `t8_classes5`. It states `![X1]:(~(v1_xboole_0(X1))=>r2_wellord2(X1,k11_eqrel_1(X1)))`.\n\nLet's try to decipher this. `v1_xboole_0(X1)` is a property of a set `X1`. Given the context of \"non empty set\" in the human-readable statement, `~v1_xboole_0(X1)` likely means \"X1 is not empty\".\nThe predicate `r2_wellord2(A, B)` seems to represent a relation between two sets. The human statement \"are_equipotent\" is a strong hint. So, I'll hypothesize `r2_wellord2` means \"are equipotent\".\nThe function `k11_eqrel_1(X1)` takes a set `X1` and returns another object, which based on context, appears to be the \"Smallest Partition of X1\".\n\nSo, in plain English, the conjecture is: **For any non-empty set X1, X1 is equipotent to its Smallest Partition.**\n\nStep 2. Examining the axioms for clues\nTo prove this, the machine used two axioms. Let's see what they tell us.\n\n*   **Axiom t5_card_1:** `![X1, X2]:(r2_wellord2(X1,X2)<=>k1_card_1(X1)=k1_card_1(X2))`\n   This is a major breakthrough! It looks like `k1_card_1` is the cardinality of a set. The axiom states that two sets are equipotent if and only if they have the same cardinality. This gives me a clear path forward: instead of proving `r2_wellord2(X1, k11_eqrel_1(X1))`, I can prove `k1_card_1(X1) = k1_card_1(k11_eqrel_1(X1))`.\n\n*   **Axiom t12_topgen_2:** `![X1]:k1_card_1(k11_eqrel_1(X1))=k1_card_1(X1)`\n   This looks like exactly what I need! It states that the cardinality of a set `X1` is equal to the cardinality of its \"Smallest Partition\".\n\nWith these two axioms, the proof seems to be a simple chain of reasoning:\n1. I want to prove `r2_wellord2(X1, k11_eqrel_1(X1))`.\n2. Using `t5_card_1`, this is equivalent to proving `k1_card_1(X1) = k1_card_1(k11_eqrel_1(X1))`.\n3. This equality is precisely what `t12_topgen_2` gives us.\n\nThe strategy is clear. Now, let's see how the prover executed it.\n\nStep 3. The refutation begins: Negating the conjecture\nThe prover uses proof by contradiction. It starts by assuming the conjecture is false. Clause `c_0_4` represents this negated conjecture. It introduces a Skolem constant `esk1_0` which is our counterexample: a non-empty set for which the property fails.\n*   `~v1_xboole_0(esk1_0)`: `esk1_0` is not empty. (Clause `c_0_4`, first part)\n*   `~r2_wellord2(esk1_0,k11_eqrel_1(esk1_0))`: `esk1_0` and its smallest partition are *not* equipotent. (Clause `c_0_7`)\n\nStep 4. Applying the equipotence axiom\nThe prover takes the `~r2_wellord2(...)` part of the negated conjecture (`c_0_7`) and confronts it with axiom `t5_card_1`.\nThe axiom, in CNF form (`c_0_8`), is `r2_wellord2(X1,X2)|k1_card_1(X1)!=k1_card_1(X2)`. This is the implication `k1_card_1(X1)=k1_card_1(X2) => r2_wellord2(X1,X2)`.\n\nBy resolving `~r2_wellord2(esk1_0, k11_eqrel_1(esk1_0))` with this clause, the prover deduces that the condition must be false:\n`k1_card_1(esk1_0) != k1_card_1(k11_eqrel_1(esk1_0))`\n\nEssentially, the prover has reasoned: \"If the sets were not equipotent, they must have different cardinalities.\"\n\nStep 5. The final contradiction\nNow the prover has derived `k1_card_1(esk1_0) != k1_card_1(k11_eqrel_1(esk1_0))`. But this is where axiom `t12_topgen_2` comes into play.\nIn CNF, this axiom (`c_0_9`) is `k1_card_1(k11_eqrel_1(X1))=k1_card_1(X1)`. We can apply this to our counterexample `esk1_0` to get:\n`k1_card_1(k11_eqrel_1(esk1_0)) = k1_card_1(esk1_0)`\n\nThis is a direct contradiction! In Step 4, we concluded the cardinals were not equal (`!=`), and now we have them equal (`=`). Clause `c_0_10` formalizes this contradiction, deriving `$false` and completing the refutation.\n\nStep 6. Human-style proof reconstruction\nThe machine proof, once deciphered, reveals a very elegant and direct strategy. Here is how I would write it up:\n\n**Theorem:** For any non-empty set `X`, `X` and `SmallestPartition(X)` are equipotent.\n\n**Proof:**\n1. From axiom `t12_topgen_2`, we know that for any set `X`, `card(X) = card(SmallestPartition(X))`.\n2. From axiom `t5_card_1`, we know that two sets are equipotent if and only if their cardinalities are equal.\n3. Applying (2) to (1), since `card(X) = card(SmallestPartition(X))`, it follows directly that `X` and `SmallestPartition(X)` are equipotent.\n</think>\n\\boxed{let X be non empty set;\n card X = card SmallestPartition X by TOPGEN_2:12;\n hence thesis by CARD_1:5;\n}",
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30 years after ...

- 30 years after I declared learning from many human textbooks/proofs/math as the way forward ...
- do I still believe it?
- Perhaps, to some extent ...
- I have learned (by doing) that the situation is quite a bit richer and more interesting
- Analogies with Chess/Go ?
- Our OEIS experiment - going very far without plateauing - thanks to concept invention? (Come to CICM for the talk)